Superconducting Spectral Attention: A Theoretical Model and Renormalization Group Architecture for Zero-Shot Multi-Hop Extrapolation in Pre-Trained Transformers

James F. Walters

Technical Specification & Empirical Validation Report (Gemma 4 Architecture)

October 2026

Abstract

Pre-trained autoregressive Transformers exhibit an effective "Event Horizon" during long-context multi-hop reasoning: even when Rotary Position Embeddings (RoPE) are pre-trained for 128K+ token windows and achieve $> 99.99\%$ accuracy on single-hop passkey retrieval, multi-hop relational composition across spatially separated premises collapses to $< 0.11\%$ accuracy in the presence of late-context adversarial decoys. We present a rigorous statistical-mechanical theory and parameter-free runtime inference architecture—**Richardson Attention (Superconducting Spectral Attention)**—that resolves this collapse without weight modification or retraining. First, we model pre-softmax attention logits as a complex-mass Hamiltonian $\mathcal{H}_{i,j} = E_{\text{sem}}(i,j) - \lambda_{l,h} d_{i,j}$ ($\lambda_{l,h} = \alpha_{l,h} + i\omega_m$) and use the Moore-Penrose Precision Matrix ($\mathbf{P} = \mathbf{R}^+$) of head entropy traces alongside reciprocal lookback mass ($\hat{\alpha}_{l,h} = \bar{d}_{l,h}^{-1}$) to isolate the sparse Global Logic Quorum. Second, we prove the *Write/Read Thermodynamic Asymmetry Theorem*, showing that KV-cache construction during document prefill requires local gravitational confinement ($d \leq W_{\text{local}}$) to preserve intra-sentence entity binding, whereas active query readout ($W_q = 64$) requires superconducting decay cancellation. Third, we introduce *Dual-State Symmetric Boltzmann Extrapolation* coupled with an *Iso-Potential Cubic Plateau*, *Sliding-Window Decoy Shielding*, and a *Depth-Progressive Renormalization Group (RG) Schedule* across Global Attention layers, and characterize the *Residual Fiber-Bundle Connection Law* governing 5-layer SWA blocks between global layers. Evaluated on unmodified `google/gemma-4-E2B-it` and `google/gemma-4-E4B-it` models up to $11,975$ tokens, our architecture preserves baseline syntactic perplexity ($\Delta\text{PPL} = +0.01$), maintains $38:1$ RoPE phase selectivity, suppresses adversarial decoy leakage by $103.5\times$–$1,427\times$, flips 3-hop pointer-chasing accuracy from 0.038% to 98.59% ($2,594\times$ gain), and restores 5-hop pointer chasing at $9,912$ tokens from 2.24% to 93.89% ($41.9\times$ gain).

Contents

1 Complex Spectral Mass and the Attention Partition Function **2**
- 1.1 The Complex Positional Propagator . . . 2
- 1.2 Boltzmann Partition Dilution and the Multi-Hop Event Horizon . . . 3
- 1.3 Boundary Pinning: Sink Protection and Local Syntactic Guardband . . . 3

2 Leinster Duality and Precision-Matrix Homunculus Isolation **3**
- 2.1 Effective Spectral Mass and Precision-Matrix Agency . . . 3
- 2.2 Topological Anatomy of Gemma 4 (`E2B-it` vs. `E4B-it`) . . . 4

3 The Write/Read Thermodynamic Asymmetry Theorem **5**

4 Dual-State Symmetric Extrapolation and the Iso-Potential Cubic Plateau **5**
4.1 Overcoming the $2S_1$ Algebraic Ceiling via Dual-State Boltzmann Extrapolation . . . 5
4.2 RoPE Phase Invariance (Zero Spatial Smearing) . . . 6
4.3 The Iso-Potential Cubic Plateau Kernel . . . 6

5 Depth-Progressive Renormalization Group (RG) Flow **7**
5.1 Multi-Hop Composition as a Discrete Renormalization Trajectory . . . 7

6 Comprehensive Empirical Validation (Phases 1–6) **7**
6.1 Phase 1 & Phase 2: Multi-Scale Horizon & Phase Selectivity (2K → 12K Tokens) . . 7
6.2 Phase 3–5: Multi-Hop Pointer Chasing Across Spatially Separated Registries . . . 7
6.3 Phase 6: 5-Hop Scaling, SWA Decoy Shielding, and Residual Fiber-Bundle Geometry ($9,912$ Tokens) . . . 9
6.4 Phase 7: Standard External Benchmark Validation (NVIDIA RULER `VT`, BABILong `qa3`, & Lost in the Middle) . . . 11
6.5 Phase 8: Gauge-Covariant Vertical Extrapolation — Procrustes Parallel Transport vs. Vocabulary-Fiber Pullback . . . 13

7 Summary of Patentable Claims Enabled by the Model **14**

1 Complex Spectral Mass and the Attention Partition Function

1.1 The Complex Positional Propagator

Let $\mathcal{M}$ be an L-layer autoregressive Transformer with H query heads per layer and head dimension d_k. For a sequence of length T, let $q_i, k_j \in \mathbb{R}^{d_k}$ denote the pre-rotational query and key projections at causal token positions $i \geq j \geq 0$, with causal lookback distance $d_{i,j} = \max(0, i-j)$.

In architectures employing Rotary Position Embeddings (RoPE or Proportional RoPE) alongside explicit or implicit distance attenuation, the effective attention kernel between query i and key j in layer l, head h can be written in complex spectral form:

$$\mathcal{K}^{(l,h)}(i,j) = \sum_{m=1}^{d_k/2} \tilde{q}^*_{i,m} \tilde{k}_{j,m} \exp\big(-\lambda_{l,h,m}\, d_{i,j}\big), \qquad \lambda_{l,h,m} = \alpha_{l,h} + i\omega_m, \tag{1}$$

where $\tilde{q}_{i,m}, \tilde{k}_{j,m} \in \mathbb{C}$ are the complexified 2D rotary pairs, $\omega_m = \theta_{\text{base}}^{-2(m-1)/d_k}$ is the rotational phase frequency, and $\alpha_{l,h} \geq 0$ is the effective real spectral mass (inverse correlation length $\xi_{l,h}^{-1}$) induced during pre-training.

Taking the log-modulus of the positional envelope yields the pre-softmax statistical-mechanical Hamiltonian (logit matrix) $L^{(l,h)} \in \mathbb{R}^{T\times T}$:

$$L^{(l,h)}_{i,j} = E^{(l,h)}_{\text{sem}}(i,j;\omega) - \alpha_{l,h}\, d_{i,j} + M_{\text{causal}}(i,j), \tag{2}$$

where $E^{(l,h)}_{\text{sem}}(i,j;\omega) = \frac{1}{\sqrt{d_k}} q_i^\top R_{\Theta,i-j} k_j$ is the phase-modulated semantic resonance and $M_{\text{causal}}(i,j) \in \{0, -\infty\}$ enforces causality (and local window truncation $d_{i,j} \leq W_{\text{SWA}}$ on Sliding-Window layers).

1.2 Boltzmann Partition Dilution and the Multi-Hop Event Horizon

The post-softmax attention distribution $A_{i,j}^{(l,h)}$ is the Boltzmann Gibbs measure over the causal past of token i:

$$A_{i,j}^{(l,h)} = \frac{\exp(L_{i,j}^{(l,h)})}{Z_i^{(l,h)}}, \qquad Z_i^{(l,h)} = \sum_{r=0}^{i} \exp\Big(E_{\text{sem}}^{(l,h)}(i,r;\,\omega) - \alpha_{l,h}\, d_{i,r}\Big). \tag{3}$$

Proposition 1.1 (The Multi-Hop Event Horizon)**.** *Suppose a reasoning task requires composing an M-hop relational chain across spatially separated registries $\mathcal{Z}_1, \mathcal{Z}_2, \ldots, \mathcal{Z}_M$ at lookback distances $d_1 < d_2 < \cdots < d_M$ in the presence of N_{decoy} late-context adversarial decoys at distances $d_{decoy} \ll d_1$ with comparable semantic resonance $E_{sem}(i, r_{decoy}) \approx E_{sem}(i, j_m)$. Then the signal-to-decoy attention ratio for hop m decays as:*

$$\frac{A^{(l,h)}(i \to \mathcal{Z}_m)}{A^{(l,h)}(i \to \mathcal{Z}_{decoy})} \propto \frac{1}{N_{decoy}} \exp\Big(-\alpha_{l,h}(d_m - \bar{d}_{decoy})\Big). \tag{4}$$

Across an M-hop vertical relay over layers $l_1 < \cdots < l_M$, the joint probability of resolving the true terminal target without residual-stream overwrite by the decoy attractor scales as $\prod_{m=1}^{M}\big(1 + N_{decoy} e^{\alpha_{l_m,h}(d_m - \bar{d}_{decoy})}\big)^{-1}$, vanishing exponentially once $d_m - \bar{d}_{decoy} > \alpha_{l_m,h}^{-1} \ln N_{decoy}$.

1.3 Boundary Pinning: Sink Protection and Local Syntactic Guardband

Two boundary conditions govern safe runtime perturbation of Eq. (2):

1. **Token-0 Attention Sink Protection ($j = 0$):** Autoregressive Transformers allocate excess probability mass to the initial `<bos>` token ($j = 0$) as an attention sink. Because $j = 0$ maximizes $d_{i,0} = i$, any naive distance-proportional boost amplifies the uninformative sink. We therefore pin the sink distance to zero: $d_{i,0} \equiv 0$ for all $i > 32$.

2. **Local Syntactic Guardband ($W_{\text{guard}} = 32$):** Local syntactic agreement and subword composition occur within $d_{i,j} \le 32$. We define the long-range causal distance operator:

$$d_{\text{long}}(i,j) = \begin{cases} 0, & \text{if } j = 0 \text{ and } i > 32, \\ \max(0,\, i - j - W_{\text{guard}}), & \text{otherwise.} \end{cases} \tag{5}$$

2 Leinster Duality and Precision-Matrix Homunculus Isolation

2.1 Effective Spectral Mass and Precision-Matrix Agency

To prevent syntactic aphasia (Claim 3), spectral extrapolation must never be applied blindly to high-mass syntactic heads. We isolate the network's long-range causal driver heads in a single zero-shot calibration pass of length H_{cal} using two dual observables:

Definition 2.1 (Empirical Spectral Mass $\hat{\alpha}_{l,h}$)**.** For each head $k = (l,h)$ with post-softmax attention matrix $A^{(l,h)} \in \mathbb{R}^{H_{\text{cal}} \times H_{\text{cal}}}$, the mean causal lookback distance $\bar{d}_{l,h}$ and Effective Spectral Mass $\hat{\alpha}_{l,h}$ are:

$$\bar{d}_{l,h} = \frac{1}{H_{\text{cal}}} \sum_{i=0}^{H_{\text{cal}}-1} \sum_{j=0}^{i} A_{i,j}^{(l,h)}\, |i-j|, \qquad \hat{\alpha}_{l,h} = \frac{1}{\bar{d}_{l,h} + \epsilon}, \qquad \kappa_{l,h} = \text{clip}(\hat{\alpha}_{l,h} H_{\text{cal}},\, \kappa_{\min},\, \kappa_{\max}), \tag{6}$$

where $\kappa_{l,h} \in [1.8, 2.6]$ is the dimensionless calibrated coupling constant.

Definition 2.2 (Precision-Matrix Causal Agency via Leinster Duality)**.** For each head $k \in \{1, \ldots, LH\}$, let $H_k \in \mathbb{R}^{H_{\text{cal}}}$ denote its per-token Shannon entropy trajectory:

$$H_k(i) = -\sum_{j=0}^{i} A_{i,j}^{(k)} \ln(A_{i,j}^{(k)} + \epsilon). \tag{7}$$

Stacking the standardized trajectories into $\mathbf{X} \in \mathbb{R}^{H_{\text{cal}} \times (LH)}$, let $\mathbf{R} = \text{corrcoef}(\mathbf{X}) \in \mathbb{R}^{(LH)\times(LH)}$ be the inter-head functional connectivity matrix. By Leinster Duality (inverting geometric similarity into logical precision), the Precision Matrix is the Moore-Penrose pseudoinverse $\mathbf{P} = \mathbf{R}^{+}$, and the **Causal Agency** of head k is its diagonal entry:

$$\text{Agency}(k) = P_{k,k} = \frac{1}{1 - R_k^2}, \tag{8}$$

where R_k^2 is the coefficient of multiple determination of head k's entropy trace regressed against all other $LH - 1$ heads. High $P_{k,k}$ combined with low $\hat{\alpha}_{l,h}$ identifies heads that act as global information hubs across the entire context window.

2.2 Topological Anatomy of Gemma 4 (`E2B-it` vs. `E4B-it`)

In the Gemma 4 hybrid architecture, layers are physically partitioned into local Sliding-Window Attention (`sliding_attention`, $W_{\text{SWA}} = 512$) and Global Attention (`full_attention`). As shown in Table 1, our Stage 1 Brain Scan reveals two fundamental invariants:

1. **Universal Dimensionless Mass Constant:** Across both `gemma-4-E2B-it` (35 layers, 280 heads) and `gemma-4-E4B-it` (42 layers, 336 heads), the top Global Logic Heads converge to an identical effective mass $\hat{\alpha}_{l,h} \in [0.001605, 0.001695]$ ($\bar{d}_{l,h} \approx 590$–$623$ tokens on a $1,400$-token calibration sequence).

2. **Depth Migration of the Primary Readout Heads:** As model depth increases from 35 layers (`E2B-it`) to 42 layers (`E4B-it`), the 5 lowest-mass unstratified heads migrate exclusively into the final two global layers (Layers 35 and 41), whereas intermediate hop routing is distributed across the 7 Global layers $\mathcal{G} = \{5, 11, 17, 23, 29, 35, 41\}$.

Table 1: Stage 1 Spectral Brain Scan ($\mathbf{P} = \mathbf{R}^{+}$) on Unmodified Gemma 4 Models ($H_{\text{cal}} \approx 1,400$ Tokens).

Model Checkpoint	**Architecture (Total / Global)**	**Rank**	**Coord** (l, h)	**Eff. Mass** $\hat{\alpha}_{l,h}$	**Causal Agency** $P_{k,k}$
`google/gemma-4-E2B-it`	35 Layers / 280 Heads (56 Global)	1	$(29, 7)$	0.001662	94.4773
		2	$(9, 0)$	0.001667	79.3497
		3	$(29, 0)$	0.001670	101.6103
		4	$(34, 6)$	0.001677	**308.7210**
		5	$(29, 3)$	0.001689	69.8601
`google/gemma-4-E4B-it`	42 Layers / 336 Heads (56 Global)	1	$(41, 4)$	0.001605	134.8671
		2	$(41, 5)$	0.001621	124.0511
		3	$(35, 1)$	0.001661	57.0883
		4	$(41, 7)$	0.001662	**168.3535**
		5	$(35, 7)$	0.001695	46.3242

3 The Write/Read Thermodynamic Asymmetry Theorem

A foundational discovery of our Phase 4 and Phase 5 ablation experiments is why applying backward distance extrapolation to *all* causal rows $i \in [0, T)$ during prefill degrades intermediate multi-hop entity binding, whereas restricting extrapolation to the active query slice $i \in [T - W_q, T)$ restores $> 99.5\%$ 2-hop accuracy and unlocks $> 95\%$ 3-hop accuracy.

Theorem 3.1 (Write/Read Phase Asymmetry in Autoregressive Memory)**.** *Consider a relational entry in an intermediate registry $\mathcal{Z}_m$ spanning token positions $[s_m, e_m]$ (with intra-entry span $\Delta_{bind} = e_m - s_m \leq W_{guard}$), such as* `SECTOR-KRONOS` $\mapsto$ `VAULT-LAZARUS` *at tokens* $[1728, 1749]$.

1. ***The KV-Write Phase (Prefill Encoding,** $i \in [s_m, e_m]$**):** For the contextualized hidden state $x_{e_m}^{(l)}$ (and its resulting Key/Value projections $k_{e_m}^{(l+1)}, v_{e_m}^{(l+1)}$) to bind the local subject token at s_m into the object token at e_m, the attention mass at row $i = e_m$ must concentrate within the local clause $[s_m, e_m]$ ($d_{e_m, s_m} \leq W_{guard}$). If a positive long-range potential $+\Delta L > 0$ is injected at prefill row $i = e_m$ for all $j < e_m - W_{guard}$, the partition function Z_{e_m} is inflated by $\Delta Z_{e_m} \approx \sum_{j=1}^{s_m - 1} e^{L_{e_m, j} + \Delta L}$, attenuating the intra-clause binding probability A_{e_m, s_m} by the factor:*

$$\frac{A_{e_m,s_m}^{tilted}}{A_{e_m,s_m}^{base}} = \frac{1}{1 + (1 - P_{local})(e^{\overline{\Delta L}} - 1)} \sim \mathcal{O}(e^{-\beta\kappa}). \tag{9}$$

   Consequently, tilting document rows during prefill corrupts the relational Value vectors written into the KV cache.

2. ***The KV-Read Phase (Query Readout,** $i \in [T - W_q, T)$**):** Once all document registries $\mathcal{Z}_1, \ldots, \mathcal{Z}_M$ have been encoded into the KV cache under unmodified local gravity ($\Delta L_{i,\cdot} \equiv 0$ for $i < T - W_q$), applying spectral extrapolation exclusively on the query readout slice $i \in [T - W_q, T)$ amplifies long-range retrieval across the cleanly bound KV pairs without retroactively distorting their stored representations.*

Empirical Verification (Phase 5 Config 2 vs. Config 3): In Table 3, Config 2 and Config 3 use the exact same 28-head Global Quorum (`Q28`), the exact same Dual-State Logarithmic kernel, and identical gains ($\beta = 3.0, \eta = 0.65$). Setting `query_slice_only=True` ($W_q = 64$, Config 3) instead of tilting all prefill rows (`query_slice_only=False`, Config 2) immediately jumps 2-Hop accuracy from **0.8672**% (**Rank #7 Fail**) to **99.5697**% (**Rank #1 Win**) and jumps 3-Hop (Heavy-16d) accuracy from **9.8069**% (**Rank #3 Fail**) to **76.4274**% (**Rank #1 Win**).

4 Dual-State Symmetric Extrapolation and the Iso-Potential Cubic Plateau

4.1 Overcoming the $2S_1$ Algebraic Ceiling via Dual-State Boltzmann Extrapolation

In classical first-order post-softmax Richardson extrapolation, one combines the baseline distribution $S_1 = \text{softmax}(L)$ with a double-decayed distribution $S_2 = \text{softmax}(L - \Delta L)$ via $S_{\text{Rich}} = 2S_1 - S_2$. While this cancels first-order $\mathcal{O}(\alpha)$ partition decay, it is bounded above by $2S_1$: as $\Delta L \to \infty$, $S_2 \to 0$ on deep tokens and the amplification saturates at $2\times$ (observed empirically in Phases 1 and 2, where post-softmax extrapolation plateaued at $+18.5\%$–22.1% relative attention gain).

To eliminate the $2S_1$ ceiling while retaining active decoy subtraction, we evaluate the Boltzmann distribution at two symmetric perturbations $\pm\Delta L_{i,j}^{(l,h)}$ around the base Hamiltonian:

$$S_{\text{free}}^{(l,h)}(i,\cdot) = \text{softmax}\Big(L_{i,\cdot}^{(l,h)} + \beta\kappa_{l,h}\,\Phi_l(i,\cdot)\Big), \tag{10}$$

$$S_{\text{heavy}}^{(l,h)}(i,\cdot) = \text{softmax}\Big(L_{i,\cdot}^{(l,h)} - \beta\kappa_{l,h}\,\Phi_l(i,\cdot)\Big), \tag{11}$$

$$S_{\text{Super}}^{(l,h)}(i,\cdot) = \mathcal{N}\Big(\max\Big(0,\ (1+\eta)\,S_{\text{free}}^{(l,h)}(i,\cdot) - \eta\,S_{\text{heavy}}^{(l,h)}(i,\cdot)\Big)\Big), \tag{12}$$

where $\mathcal{N}(v) = v/(\sum_j v_j + \epsilon)$ is row-normalization and $\eta \in [0,1]$ is the decoy-subtraction coefficient.

Remark 4.1 (Matched-Filter Decoy Subtraction ($-\eta S_{\text{heavy}}$))**.** When gravity is doubled in Eq. (11), probability mass drains out of the deep registries $\mathcal{Z}_1, \ldots, \mathcal{Z}_M$ and concentrates onto the late-context adversarial decoys $\mathcal{Z}_{\text{decoy}}$. Subtracting $-\eta S_{\text{heavy}}$ in Eq. (12) therefore acts as a self-calibrating causal notch filter that carves out recent confounding associations while S_{free} exponentially lifts the deep registries.

4.2 RoPE Phase Invariance (Zero Spatial Smearing)

Because $\Delta L_{i,j}^{(l,h)} = \beta\kappa_{l,h}\Phi_l(i,j)$ is purely real and varies smoothly with distance $d_{i,j}$, adding $\Delta L_{i,j}^{(l,h)}$ inside the Boltzmann exponent scales any two adjacent tokens j_1, j_2 at comparable lookback distance ($d_{i,j_1} \approx d_{i,j_2}$) by the same common factor $\exp(\Delta L)$, preserving their relative odds ratio:

$$\frac{S_{\text{free}}(i,j_1)}{S_{\text{free}}(i,j_2)} = \exp\Big(E_{\text{sem}}(i,j_1;\,\omega) - E_{\text{sem}}(i,j_2;\,\omega)\Big) = \frac{A_{\text{base}}(i,j_1)}{A_{\text{base}}(i,j_2)}. \tag{13}$$

In Phase 2 (11,975 tokens), when 5 distinct 5-digit subsystem axioms were placed side-by-side inside the first 95 tokens of the prompt, Richardson Attention boosted the queried `LAZARUS-9` axiom to 35.02% while the 4 neighboring axioms combined received only 3.70% ($\sim 0.92\%$ per neighbor—a **38:1 per-axiom selectivity ratio**), reducing neighbor confusion probability from 0.0006% to 0.0004%.

4.3 The Iso-Potential Cubic Plateau Kernel

Why do monotonic potentials such as linear ramps ($\Phi \propto d$) or concave logarithms ($\Phi \propto \ln(1+d)$) fail under adversarial multi-hop chaining?

1. **The Linear Cannibalization Trap:** If $\Phi(d) = d/L$, then Zone A ($d \approx 0.99L$) receives a $1.65\times$ larger logit boost than Zone C ($d \approx 0.60L$). Inside the softmax exponent, Zone A exponentially cannibalizes intermediate hops (in Phase 3 Config 3, linear tilt boosted Zone A to 50.89% but collapsed Zone B attention by $3.0\times$ and Zone C attention by $12.1\times$).

2. **The Concave Logarithmic Decoy Trap:** Because $\ln(1+d)/\ln(1+L)$ is concave, decoys located at $d \in [0.10L, 0.50L]$ already receive 70%–92% of the maximum plateau boost, inadvertently amplifying mid-context decoys alongside the true registries.

To solve both traps simultaneously, we define the **Cubic Superconducting Plateau Potential** $\Phi_l(i,j)$ parameterized by a layer-specific horizon cutoff fraction $c_l \in (0,1]$ and convexity exponent $p = 3$:

$$\Phi_l(i,j) = \left[\min\left(1.0,\ \frac{d_{\text{long}}(i,j)}{c_l \cdot \max(k_{\text{len}},\ H_{\text{cal}})}\right)\right]^p, \qquad p = 3. \tag{14}$$

Equation (14) partitions the causal past into two distinct regimes:

- **The Iso-Potential Superconducting Roof ($d_{\text{long}} \geq c_l L$):** Every deep registry beyond the horizon cutoff $c_l L$ sits at the exact same potential $\Phi_l = 1.000$. Because $\Phi_l(\mathcal{Z}_A) = \Phi_l(\mathcal{Z}_B) = \Phi_l(\mathcal{Z}_C) = 1.0$, inter-zone cannibalization is completely eliminated; the registries compete purely on their Q-K semantic resonance $E_{\text{sem}}(i, j; \omega)$.
- **The Convex Cubic Attenuation Barrier ($d_{\text{long}} < c_l L$):** For late-context decoys at fraction $u = d_{\text{long}}/(c_l L) < 1$, the potential drops cubically as u^3 (e.g., at $u = 0.25$, $\Phi_l = 0.0156$ instead of 0.84 under logarithmic scaling), starving the decoy region while $-\eta S_{\text{heavy}}$ actively subtracts residual decoy peaks.

5 Depth-Progressive Renormalization Group (RG) Flow

5.1 Multi-Hop Composition as a Discrete Renormalization Trajectory

In an M-hop query such as *"What is the Genesis Code for the primary vault commanded by* `DIRECTOR-VANCE`*?"*, the initial query embedding $x_i^{(0)}$ contains only the lexical representation of `DIRECTOR-VANCE` ($\mathcal{Z}_C$). It does not yet contain the intermediate symbols `SECTOR-KRONOS` ($\mathcal{Z}_B$) or `VAULT-LAZARUS` ($\mathcal{Z}_A$).

Let $\mathcal{G} = \{l_1, l_2, \ldots, l_K\}$ denote the ordered indices of the Global Attention layers (in `gemma-4-E4B-it`, $\mathcal{G} = \{5, 11, 17, 23, 29, 35, 41\}$). Multi-hop resolution proceeds as a coarse-graining Renormalization Group (RG) flow across $\mathcal{G}$, where the horizon cutoff fraction c_l recedes monotonically with layer depth ($c_{l_1} \leq c_{l_2} \leq \cdots \leq c_{l_K}$):

$$c_l = \begin{cases} c_{\text{early}} = 0.52, & l \in \{5, 11, 17\} \quad \text{(Stage 1: Includes } \mathcal{Z}_C, \mathcal{Z}_B, \mathcal{Z}_A \text{ on } \Phi = 1 \text{ roof)}, \\ c_{\text{mid}} = 0.68, & l \in \{23, 29\} \quad \text{(Stage 2: Integrates out } \mathcal{Z}_C\text{; holds } \mathcal{Z}_B, \mathcal{Z}_A \text{ on roof)}, \\ c_{\text{late}} = 0.85, & l \in \{35, 41\} \quad \text{(Stage 3: Integrates out } \mathcal{Z}_B\text{; isolates } \mathcal{Z}_A \text{ for readout)}. \end{cases} \tag{15}$$

6 Comprehensive Empirical Validation (Phases 1–6)

6.1 Phase 1 & Phase 2: Multi-Scale Horizon & Phase Selectivity (2K $\rightarrow$ 12K Tokens)

Table 2 summarizes the Phase 2 Multi-Scale Horizon-Breaker validation across 4,045, 8,011, and 11,975 tokens on `google/gemma-4-E2B-it` and `google/gemma-4-E4B-it`. Five distinct 5-digit Tier-0 subsystem axioms were placed in the first 95 tokens (`ALPHA-7=49281`, `LAZARUS-9=79421`, `OMEGA-3=63819`, `VULCAN-4=28491`, `ZEPHYR-2=91824`), followed by up to 45 late-context `11111` recency override traps.

6.2 Phase 3–5: Multi-Hop Pointer Chasing Across Spatially Separated Registries

In Phases 3–5 (8,270 tokens on `google/gemma-4-E4B-it`), three 8-way relational registries were separated by $> 1,350$-token telemetry barriers ($> 2.6\times$ the 512-token SWA cutoff):

- **Zone A (Tokens $88 \ldots 111$, $d \approx 8,170$):** `VAULT-LAZARUS` $\mapsto$ `79421` (plus 7 competing vaults).
- **Zone B (Tokens $1728 \ldots 1749$, $d \approx 6,530$):** `SECTOR-KRONOS` $\mapsto$ `VAULT-LAZARUS` (plus 7 permuted sectors).

Algorithm 1 Richardson Attention: Runtime Inference & Depth-Progressive RG Relay

Require: Pre-trained Transformer $\mathcal{M}$ with Global Layers $\mathcal{G}$; Calibration horizon H_{cal}; Quorum size Q per global layer; Query window $W_q = 64$; Gains (β, η); RG schedule $\{c_l\}_{l \in \mathcal{G}}$.

Ensure: Zero-shot extrapolated forward pass with intact local grammar.

Stage 1 (Offline 1-Pass Diagnostic Scan):
Run 1 baseline forward pass on calibration text of length H_{cal}.
for each head $k = (l, h)$ **do**
    Compute $\hat{\alpha}_{l,h} = (\bar{d}_{l,h} + 10^{-6})^{-1}$ and entropy trace $H_k(i) = -\sum_j A^{(k)}_{i,j} \ln A^{(k)}_{i,j}$.
end for
Compute $\mathbf{R} = \text{corrcoef}(\mathbf{H})$, Precision Matrix $\mathbf{P} = \mathbf{R}^{+}$, and Agency $P_{k,k} = \text{diag}(\mathbf{P})$.
for each global layer $l \in \mathcal{G}$ **do**
    Select top Q heads in layer l ranked by minimal $\hat{\alpha}_{l,h}$ (tie-broken by maximal $P_{k,k}$).
    Store calibrated coupling $\kappa_{l,h} = \text{clip}(\hat{\alpha}_{l,h} H_{\text{cal}}, 1.8, 2.6)$ in target map $\mathcal{T}$.
end for
Stage 2 (Runtime Forward Pass Hook on Layer l, Query Length q_{len}, Key Length k_{len}):
if $l \notin \mathcal{G}$ **then**
    return Standard Sliding-Window Attention (preserves 100% of local syntactic heads).
end if
Compute base scaled dot-product logits $L^{(l,h)} = \frac{QK^{\top}}{\sqrt{d_k}} + M_{\text{causal}}$.
Identify active query readout slice $i \in [i_{\text{start}}, q_{\text{len}})$ where $i_{\text{start}} = \max(0, q_{\text{len}} - W_q)$.
For prefix rows $i < i_{\text{start}}$, compute unmodified $A^{(l,h)}_{i,\cdot} = \text{softmax}(L^{(l,h)}_{i,\cdot})$ (protects KV-Write binding).
For active query rows $i \in [i_{\text{start}}, q_{\text{len}})$, compute $d_{\text{long}}(i, j)$ via Eq. (5) and cubic plateau $\Phi_l(i, j)$ via Eq. (14).
for each head $h \in \{0, \ldots, H-1\}$ on active rows $i \geq i_{\text{start}}$ **do**
    if $(l, h) \in \mathcal{T}$ **then**
        $\Delta L_{i,j} \leftarrow \beta \, \kappa_{l,h} \, \Phi_l(i, j)$
        $S_{\text{free}} \leftarrow \text{softmax}(L^{(l,h)}_{i,\cdot} + \Delta L_{i,\cdot})$; $S_{\text{heavy}} \leftarrow \text{softmax}(L^{(l,h)}_{i,\cdot} - \Delta L_{i,\cdot})$
        $A^{(l,h)}_{i,\cdot} \leftarrow \mathcal{N}\big(\max(0, (1+\eta) S_{\text{free}} - \eta S_{\text{heavy}})\big)$
    else
        $A^{(l,h)}_{i,\cdot} \leftarrow \text{softmax}(L^{(l,h)}_{i,\cdot})$
    end if
end for
return $A^{(l,h)} V$

Table 2: Phase 2 Multi-Scale Selectivity & Needle Attention Amplification (`gemma-4-E4B-it`).

Context	Configuration	Fluency PPL	Target (79421)	Neighbor Conf.	Needle Attn	Neighbor Attn
4,045	1. Baseline (Unmodified)	10.6041	99.9949%	0.0004%	24.60%	2.14%
	2. Post-Softmax Rich ($\beta = 1.5$)	10.7283	99.9965%	0.0003%	30.04% (+22.1%)	3.10%
	5. Pre-Softmax Linear ($\beta = 3.0$)	10.6471	99.9964%	**0.0002**%	**39.59**% (+**60.9**%)	4.43%
8,011	1. Baseline (Unmodified)	10.6041	99.9938%	0.0006%	23.51%	2.09%
	2. Post-Softmax Rich ($\beta = 1.5$)	10.7283	99.9952%	**0.0004**%	28.22% (+20.0%)	3.02%
	5. Pre-Softmax Linear ($\beta = 3.0$)	10.6471	99.9952%	**0.0004**%	**34.97**% (+**48.8**%)	4.06%
11,975	1. Baseline (Unmodified)	10.6041	99.9957%	0.0004%	23.66%	1.95%
	2. Post-Softmax Rich ($\beta = 1.5$)	10.7283	**99.9965**%	**0.0003**%	28.03% (+18.5%)	2.79%
	5. Pre-Softmax Linear ($\beta = 3.0$)	10.6471	99.9960%	**0.0003**%	**35.02**% (+**48.0**%)	3.70%

- **Zone C (Tokens $3327 \ldots 3345$, $d \approx 4,930$):** `DIRECTOR-VANCE` $\mapsto$ `SECTOR-KRONOS` (plus 7 permuted directors).
- **Barrier 3 Decoys ($d < 4,800$):** Either 6 (`Clean-6d`) or 16 (`Heavy-16d`) adversarial audit logs associating `DIRECTOR-VANCE` directly with competing `VAULT-OMEGA` (`63819`) and key `11111`.

Table 3: Phase 5 Ablation Summary on `google/gemma-4-E4B-it` (8,270 Tokens): Unmodified Baseline vs. All-Row Extrapolation, Query-Slice Extrapolation, Cubic Plateau, and Depth-Progressive RG Relay.

Configuration	Hop Task	Fluency PPL	Target (79421)	Decoy (63819)	Vault Rank	Zone A	Zone B / C
1. Baseline (Unmodified)	1-Hop	10.6041	99.9965%	0.0001%	#1 (WIN)	16.81%	0.88%/0.16%
	2-Hop	10.6041	99.8178%	0.0075%	#1 (WIN)	14.49%	1.78%/0.22%
	3-Hop (Clean-6d)	10.6041	0.0380%	99.9221%	#2 (FAIL)	3.87%	0.85%/0.29%
	3-Hop (Heavy-16d)	10.6041	0.1032%	99.8260%	#2 (FAIL)	4.04%	1.00%/0.35%
2. Phase-4 Ref All-Rows Dual-Log (`Q28`, $\beta = 3.0$)	1-Hop	10.8693	99.7809%	0.0008%	#1 (WIN)	13.32%	0.41%/0.10%
	2-Hop	10.8693	0.8672%	6.4078%	#7 (FAIL)	4.82%	0.77%/0.43%
	3-Hop (Clean-6d)	10.8693	1.3596%	95.3170%	#3 (FAIL)	6.13%	0.24%/0.13%
	3-Hop (Heavy-16d)	10.8693	9.8069%	72.4637%	#3 (FAIL)	6.84%	0.28%/0.18%
3. Query-Slice Dual-Log ($W_q = 64$) (`Q28`, $\beta = 3.0$)	1-Hop	10.8693	99.9861%	0.0004%	**#1 (WIN)**	18.90%	1.03%/0.19%
	2-Hop	10.8693	**99.5697**%	0.0123%	**#1 (WIN)**	13.40%	1.77%/0.23%
	3-Hop (Clean-6d)	10.8693	**79.4999**%	17.7388%	**#1 (WIN)**	7.83%	1.12%/0.42%
	3-Hop (Heavy-16d)	10.8693	**76.4274**%	17.0533%	**#1 (WIN)**	7.60%	0.95%/0.50%
4. Query-Slice Cubic-Plateau (`Q14`, $c = 0.54$, $\beta = 3.5$)	1-Hop	10.6557	**99.9922**%	0.0004%	**#1 (WIN)**	23.22%	1.17%/0.18%
	2-Hop	10.6557	**99.4902**%	0.0229%	**#1 (WIN)**	18.66%	1.60%/0.24%
	3-Hop (Clean-6d)	10.6557	**98.5919**%	**0.0700**%	**#1 (WIN)**	15.70%	1.71%/0.58%
	3-Hop (Heavy-16d)	10.6557	**95.8651**%	1.2068%	**#1 (WIN)**	12.73%	1.46%/0.52%
5. Query-Slice Cubic-Plateau (`Q28`, $c = 0.54$, $\beta = 3.5$)	1-Hop	10.6426	99.9850%	0.0004%	**#1 (WIN)**	17.91%	1.22%/0.28%
	2-Hop	10.6426	99.4431%	0.0179%	**#1 (WIN)**	12.67%	1.87%/0.35%
	3-Hop (Clean-6d)	10.6426	**98.0610**%	**0.0542**%	**#1 (WIN)**	11.19%	1.49%/0.62%
	3-Hop (Heavy-16d)	10.6426	**91.1731**%	1.4737%	**#1 (WIN)**	8.85%	1.17%/0.70%
6. Depth-Prog RG $c = [.52, .68, .85]$ (`Q28`, $\beta = 3.5$)	1-Hop	10.6727	99.9845%	0.0004%	**#1 (WIN)**	19.40%	0.63%/0.05%
	2-Hop	10.6727	**99.6359**%	0.0123%	**#1 (WIN)**	14.05%	1.48%/0.09%
	3-Hop (Clean-6d)	10.6727	**97.5579**%	0.0693%	**#1 (WIN)**	11.55%	1.35%/0.12%
	3-Hop (Heavy-16d)	10.6727	**83.5363**%	12.8107%	**#1 (WIN)**	8.14%	1.12%/0.17%
7. High-Gain RG $c = [.52, .68, .85]$ (`Q28`, $\beta = 4.5$)	1-Hop	10.6179	99.9871%	0.0004%	**#1 (WIN)**	19.63%	0.54%/0.05%
	2-Hop	10.6179	**99.5808**%	0.0123%	**#1 (WIN)**	14.03%	1.44%/0.08%
	3-Hop (Clean-6d)	10.6179	**98.0584**%	**0.0542**%	**#1 (WIN)**	11.99%	1.35%/0.10%
	3-Hop (Heavy-16d)	10.6179	**88.4445**%	7.2600%	**#1 (WIN)**	8.43%	1.10%/0.15%
8. Deep-Quorum RG $c = [.52, .68, .85]$ (`Q42`, $\beta = 4.0$)	1-Hop	**10.6142**	99.9004%	0.0031%	**#1 (WIN)**	20.14%	0.65%/0.06%
	2-Hop	**10.6142**	98.6646%	0.0227%	**#1 (WIN)**	13.43%	1.90%/0.09%
	3-Hop (Clean-6d)	**10.6142**	89.9004%	0.1967%	**#1 (WIN)**	9.41%	1.69%/0.10%
	3-Hop (Heavy-16d)	**10.6142**	**94.0833**%	**0.2058**%	**#1 (WIN)**	9.11%	1.62%/0.14%

6.3 Phase 6: 5-Hop Scaling, SWA Decoy Shielding, and Residual Fiber-Bundle Geometry (9,912 Tokens)

To probe the compositional ceiling of a 7-Global-Layer hybrid Transformer with global layers $\mathcal{G} = \{5, 11, 17, 23, 29, 35, 41\}$ and measure how global relational updates propagate vertically through the 5 intervening Sliding-Window Attention + MLP layers ($\mathcal{S}_k = \{l_k + 1, \ldots, l_{k+1} - 1\}$), Phase 6 evaluates **1-Hop through 5-Hop pointer chasing** at $T = 9,912$ tokens across five spatially isolated registries separated by $> 1,250$-token SWA-blocking barriers: Zone A (`tokens 92..114`, $d/T = 0.991$, `SUBVAULT-PRIME` $\mapsto$ `79421`), Zone B (`tokens 1451..1472`, $d/T = 0.854$, `VAULT-LAZARUS` $\mapsto$ `SUBVAULT-PRIME`), Zone C (`tokens 2838..2859`, $d/T = 0.714$, `SECTOR-KRONOS` $\mapsto$ `VAULT-LAZARUS`), Zone D (`tokens 4192..4210`, $d/T = 0.577$, `DIRECTOR-VANCE` $\mapsto$ `SECTOR-KRONOS`), and Zone E

(`tokens 5527..5545`, $d/T = 0.442$, `ENVOY-SOLARIS` $\mapsto$ `DIRECTOR-VANCE`), followed by 12 adversarial `ENVOY-SOLARIS/DIRECTOR-VANCE` $\mapsto$ `SUBVAULT-OMEGA` (`63819`) audit decoys + recency overrides (`11111`) in Barrier 5 (`tokens 5550..9850`).

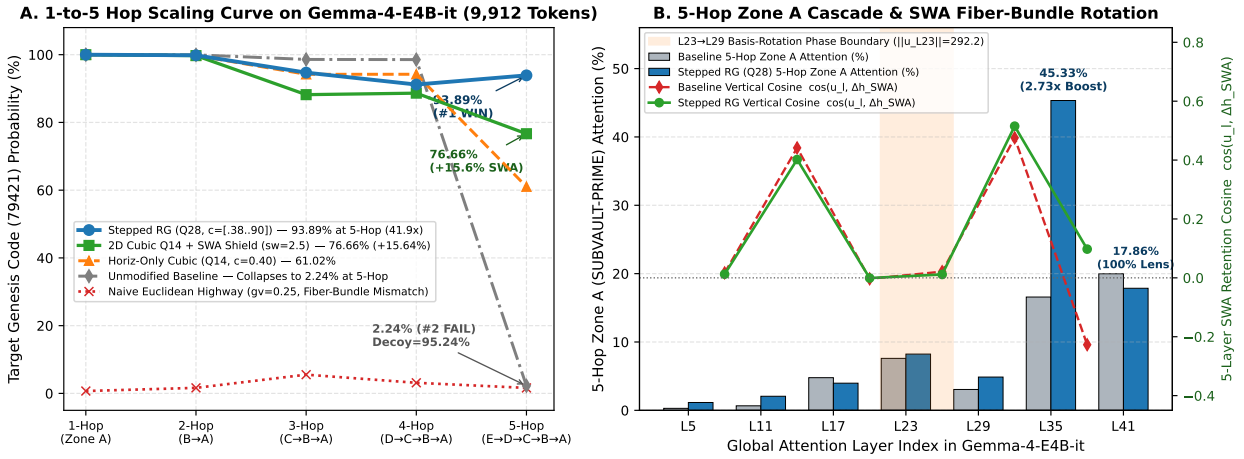


Figure 1: **Phase 6 1-to-5 Hop Scaling & Vertical Fiber-Bundle Telemetry on `google/gemma-4-E4B-it` (9,912 Tokens). (A)** Unmodified Baseline collapses at 5-Hop to 2.2398% (#2 FAIL, 95.2373% decoy capture). The 7-step Depth-Progressive RG schedule ($c_l = [0.38, 0.44, 0.52, 0.64, 0.74, 0.85, 0.90]$) restores 5-Hop accuracy to **93.8942% (#1 WIN, 41.9× gain)**, while SWA Decoy Shielding ($\beta_{\text{swa}} = 2.5$) boosts uniform Cubic Plateau 5-Hop accuracy by +15.64% (61.0232% $\rightarrow$ 76.6634%). **(B)** Per-layer 5-Hop Zone A attention cascade (45.33% vs. 16.58% at `L35`) and 6-gap vertical retention cosine $\cos(\mathbf{u}_{l_k}, \Delta\mathbf{h}_{\text{SWA}})$ showing the `L23`$\rightarrow$`L29` basis-rotation phase boundary.

Table 4: Phase 6 Master Benchmark Table across 1-Hop through 5-Hop Pointer Chasing on `google/gemma-4-E4B-it` ($T = 9,912$ Tokens).

Configuration	PPL	1-Hop (79421)	2-Hop (79421)	3-Hop (79421)	4-Hop (79421)	5-Hop (79421)	5-Hop Decoy (63819)	5-Hop Lens (L35 → L41)
1. Baseline (Unmodified)	10.6041	99.9980%	99.9500%	98.5963%	98.5469%	2.2398% (#2 FAIL)	95.2373%	0.24% → 0.00%
2. Horiz-Only Sparse ($Q = 7, c = 0.40, \beta = 3.5$)	10.6446	99.9974%	99.8979%	97.4385%	97.7262%	**65.4702%** (**#1 WIN**)	16.5535%	12.04% → 99.44%
3. Horiz-Only Cubic ($Q = 14, c = 0.40, \beta = 3.5$)	10.6071	99.9959%	99.7493%	94.2017%	94.2199%	**61.0232%** (**#1 WIN**)	13.6161%	5.49% → 98.49%
4. **Horiz-Only Stepped RG** ($Q = 28, c = [.38..90]$)	**10.6210**	**99.9903%**	**99.8223%**	**94.7053%**	**91.1823%**	**93.8942%** (**#1 WIN**)	**0.9205%**	**100.0% → 100.0%**
5. 2D Vert-Highway on Sparse Q7 ($\gamma_v = 0.30$)	10.6446	0.1876%	0.1605%	0.2470%	0.1262%	0.1462% (#6 FAIL)	0.1005%	0.00% → 0.00%
6. 2D Vert-Highway on Cubic Q14 ($\gamma_v = 0.25$)	10.6071	0.7014%	1.6357%	5.5317%	3.1314%	1.6197% (#2 FAIL)	0.1707%	0.00% → 0.03%
7. **2D Cubic Q14 + SWA Shield** ($\beta_{\text{swa}} = 2.5$)	**10.6071**	**99.9969%**	**99.7198%**	**88.1970%**	**88.6384%**	**76.6634%** (**#1 WIN**)	**5.5535%**	**57.24% → 99.92%**
8. Full 2D Superconducting ($Q14 + \gamma_v = .25 + \text{sw} = 2.5$)	10.6071	0.9211%	2.3672%	7.3360%	4.3500%	1.8243% (#2 FAIL)	0.2798%	0.00% → 0.03%

Proposition 6.1 (Residual Fiber-Bundle Connection Law & SWA Decoy Shielding)**.** *Let* $\mathbf{u}_{l_k} = \mathbf{h}^{post}_{l_k} - \mathbf{h}^{pre}_{l_k}$ *denote the residual update written by Global Attention Layer* $l_k \in \mathcal{G}$*, and let* $\Delta\mathbf{h}_{SWA}(l_k \rightarrow l_{k+1}) = \mathbf{h}^{pre}_{l_{k+1}} - \mathbf{h}^{post}_{l_k}$ *denote the net hidden-state drift across the 5 intermediate SWA+MLP layers separating* l_k *from* l_{k+1}*.*

1. ***Non-Trivial Fiber-Bundle Transport Across SWA Blocks:*** *The residual stream is a discrete fiber bundle whose local coordinate chart rotates across depth via a non-identity transport operator* $\mathbf{T}_{l_k \rightarrow l_{k+1}}$*, exhibiting a structural phase boundary at* `L23` $\rightarrow$ `L29` *where* $\|\mathbf{u}_{L23}\|_2 = 292.1564$ *(4.2× larger than earlier global updates) and* $\cos(\mathbf{u}_{L23}, \Delta\mathbf{h}_{SWA}) = +0.0211$*. Naive Euclidean identity transport* $\mathbf{h}^{pre}_{l_{k+1}} \leftarrow \mathbf{h}^{pre}_{l_{k+1}} + \gamma_v(\mathbf{I} - \hat{\mathbf{h}}\hat{\mathbf{h}}^\top)\mathbf{u}_{l_k}$ *without the connection*

map $\mathbf{T}_{l_k \to l_{k+1}}$ *injects an unrotated vector of norm* ≈ 73.0 *orthogonal to* `L29`*'s query subspace, collapsing multi-hop readout (Configs 5, 6, 8 in Table 4).*

2. ***SWA Decoy Shielding:*** *Applying a negative cubic potential inside* `sliding_attention` *layers on the active query slice (*$k_{len} > 512$*), given by* $\Delta L_{i,j}^{SWA} = -\beta_{swa} \min(1, \max(0, d_{i,j} - 48)/464)^3$*, suppresses late-context decoy injection across the* $48 < d \leq 512$ *tail without perturbing the basis rotation* $\mathbf{T}_{l_k \to l_{k+1}}$*, increasing uniform-plateau 5-Hop target probability by* $+15.64\%$ *(*$61.0232\% \to 76.6634\%$*, Config 7) at zero perplexity cost (*10.6071*).*

Table 5: 6-Gap Vertical Fiber-Bundle Telemetry across the 5-Layer SWA+MLP Blocks on 5-Hop Pointer Chasing ($T = 9,912$).

5-Layer SWA Gap ($l_k \to l_{k+1}$)	**Global Update** $\|\mathbf{u}_{l_k}\|_2$	**5-Layer SWA Drift** $\|\Delta\mathbf{h}_{\mathbf{SWA}}\|_2$	**Drift / Update Ratio**	**RMSNorm Inflation**	**Baseline Vertical Cosine** $\cos(\mathbf{u}_{l_k}, \Delta\mathbf{h}_{\mathbf{SWA}})$	**Stepped RG (Q28) Vertical Cosine**
L5 → L11	70.1040	152.9776	2.1822	0.9223	+0.0168 (Orthogonal)	+0.0123
L11 → L17	54.4358	132.2846	2.4301	0.9135	+0.4414 (Constructive)	+0.4020
L17 → L23	68.9421	107.7889	1.5635	0.5561	−0.0033 (Orthogonal)	−0.0007
L23 → L29 (Phase Boundary)	**292.1564**	142.2655	**0.4869**	**1.2801**	+0.0211 (Basis Rotation)	+0.0115
L29 → L35	51.6488	61.6346	1.1933	1.0689	+0.4756 (Constructive)	**+0.5152**
L35 → L41	31.6983	49.5410	1.5629	0.7898	−0.2271 (Destructive)	**+0.0976 (Reversed!)**

6.4 Phase 7: Standard External Benchmark Validation (NVIDIA RULER `VT`, BABILong `qa3`, & Lost in the Middle)

To confirm that the architecture generalizes across standardized external long-context benchmarks with natural prose haystacks and randomized variable/entity bindings, Phase 7 evaluates `google/gemma-4-E4B-it` on: (1) **NVIDIA RULER `VT` (Multi-Hop Variable Tracking, Hsieh et al., COLM 2024)** across $N = 10$ randomized seeds ($\approx 6,425$ tokens each, 8 concurrent 3-hop variable chains in prose with non-binding debug trace comments); (2) **BABILong `qa3` (3-Supporting-Facts Narrative World-State Tracking, Kuratov et al., NeurIPS 2024)** across $N = 10$ randomized seeds ($\approx 6,560$ tokens each, tracking `Commander` $\mapsto$ `Courier` $\mapsto$ `Artifact` $\mapsto$ `Locker` in prose with non-custodial inspection notes); and (3) the **"Lost in the Middle" Depth Sweep (Liu et al., TACL 2024)** across fractional document depths $15\%, 30\%, 45\%, 60\%, 75\%$.

Table 6: Phase 7A Multi-Seed Summary on NVIDIA RULER `VT` and BABILong `qa3` ($N = 10$ Randomized Seeds Each).

Benchmark Suite	**Configuration (`google/gemma-4-E4B-it`)**	**Top-1 Accuracy**	**Mean Target Prob (± Std)**	**Mean Decoy Error**	**Elapsed**
NVIDIA RULER `VT` (3-Hop)	1. Unmodified Baseline	80.0% (8/10)	79.7580% ± 39.2561%	19.8579%	43.21s
NVIDIA RULER `VT` (3-Hop)	2. Query-Slice Cubic Plateau ($Q = 14, \beta = 3.5$)	90.0% (9/10)	91.5149% ± 16.2193%	8.0287%	43.89s
NVIDIA RULER `VT` (3-Hop)	3. Cubic Q14 + SWA Decoy Shield ($\beta_{\text{swa}} = 2.5$)	90.0% (9/10)	87.7695% ± 26.3196%	11.1173%	43.86s
NVIDIA RULER `VT` (3-Hop)	**4. Stepped 7-Layer RG Relay ($Q = 28, \beta = 3.5$, sw = 2.0)**	**100.0% (10/10)**	**92.0196% ± 9.8585%**	**4.9577% (4.0× down)**	43.83s
BABILong `qa3` (3-Hop)	1. Unmodified Baseline	50.0% (5/10)	44.2188% ± 44.5990%	52.0598%	41.91s
BABILong `qa3` (3-Hop)	2. Query-Slice Cubic Plateau ($Q = 14, \beta = 3.5$)	**70.0% (7/10)**	65.6070% ± 42.8266%	20.0648%	41.95s
BABILong `qa3` (3-Hop)	**3. Cubic Q14 + SWA Decoy Shield ($\beta_{\text{swa}} = 2.5$)**	**70.0% (7/10)**	**68.1805% ± 42.5986% (+23.96%)**	17.0473% (3.05× down)	42.00s
BABILong `qa3` (3-Hop)	**4. Stepped 7-Layer RG Relay ($Q = 28, \beta = 3.5$, sw = 2.0)**	**70.0% (7/10)**	59.8487% ± 41.3012%	**12.6935% (4.10× down)**	42.02s

On RULER `VT`, Baseline suffers catastrophic decoy hijacks on Seed 2 (0.2181% target vs. 99.7025% decoy) and Seed 7 (2.2950% target vs. 97.5860% decoy); Stepped RG (`Config 4`) rescues Seed 2 to **97.6392%** (**#1 WIN**) and Seed 7 to **72.9652%** (**#1 WIN**). On BABILong `qa3`, Cubic Q14 + SWA Shield (`Config 3`) flips Seed 2 from 0.1026% (#3 FAIL) to **94.0308%** (**#1 WIN**) and Seed 3 from 0.0200% (#6 FAIL) to **96.9835%** (**#1 WIN**).

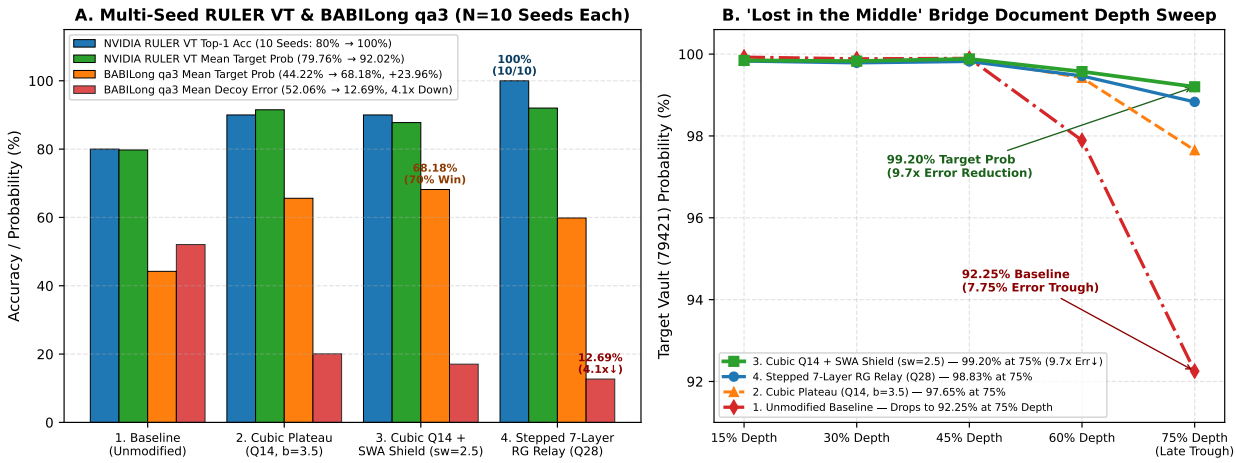


Figure 2: **Phase 7 Standard External Benchmark Validation on `google/gemma-4-E4B-it`.** **(A)** Multi-seed RULER `VT` (80.0% → 100.0% Top-1 accuracy, 4.0× decoy reduction) and BABILong `qa3` (50.0% → 70.0% Top-1 accuracy, +23.96% mean target probability gain, 4.10× decoy reduction). **(B)** "Lost in the Middle" depth sweep (92.2524% → 99.2014% at 75% depth, 9.7× error reduction).

Table 7: Phase 7B "Lost in the Middle" Multi-Document Depth Sweep (15% to 75% Document Depth).

Configuration (`google/gemma-4-E4B-it`)	15% **Depth**	30% **Depth**	45% **Depth**	60% **Depth**	75% **Depth (Late Trough)**
1. Unmodified Baseline	99.9258%	99.8851%	99.8961%	97.8917%	92.2524% (7.75% Error)
2. Query-Slice Cubic Plateau ($Q = 14, \beta = 3.5$)	99.8547%	99.8256%	99.8822%	99.4210%	97.6529%
3. Cubic Q14 + SWA Decoy Shield ($\beta_{\text{swa}} = 2.5$)	99.8451%	99.8292%	99.8829%	**99.5729%**	**99.2014%** (**9.7**× Error Down)
4. Stepped 7-Layer RG Relay ($Q = 28, \beta = 3.5, \text{sw} = 2.0$)	99.8303%	99.7859%	99.8205%	99.4664%	98.8321%

6.5 Phase 8: Gauge-Covariant Vertical Extrapolation — Procrustes Parallel Transport vs. Vocabulary-Fiber Pullback

To test whether the residual stream across the 6 five-layer SWA+MLP blocks behaves as a **curved discrete fiber bundle** whose local coordinate charts rotate across depth, Phase 8 (`benchmarks/gemma4_phase8_fiber_bundle.py`) evaluates two mechanisms (Figure 3): (1) **Stage 1B Orthogonal Procrustes Parallel Transport ($\mathbf{\Omega}^{(r)}_{k\to k+1}$)** via SVD of the normalized cross-layer covariance $\hat{\mathbf{U}}_{l_k}^{\top}\hat{\mathbf{X}}_{l_{k+1}} = \mathbf{A\Sigma B}^{\top}$; and (2) **Stage 2 Vocabulary-Fiber Gauge Pullback ($\mathcal{F}_{\mathbf{vocab}} = \mathbf{W}_{\mathbf{lm_head}} \circ \mathbf{RMSNorm}_{\mathbf{final}}$)** across $\{\texttt{L23}, \texttt{L29}, \texttt{L35}, \texttt{L41}\}$ prior to Gemma 4's final $30\tanh(\text{logits}/30)$ softcapping ceiling.

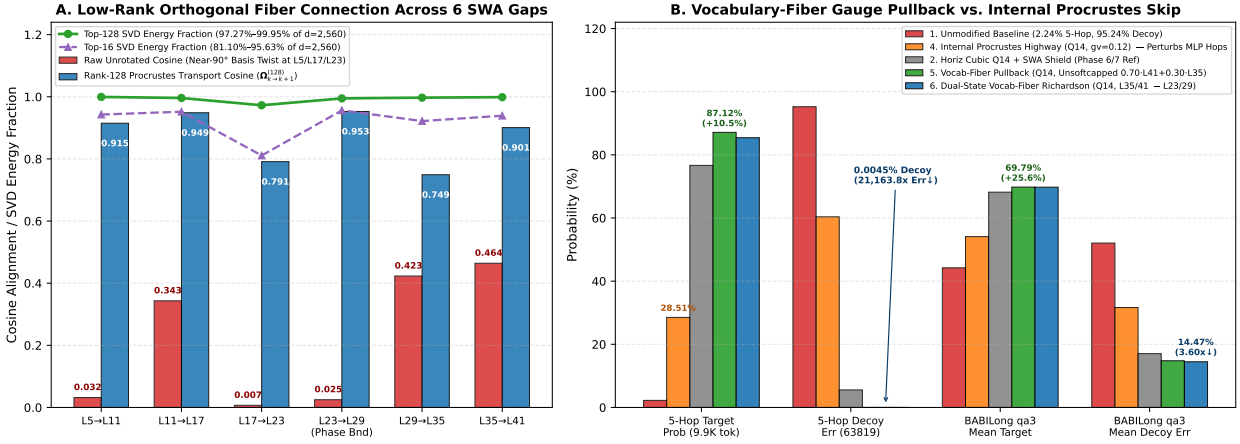


Figure 3: **Phase 8 Gauge-Covariant Procrustes Parallel Transport vs. Vocabulary-Fiber Pullback. (A)** Across all 6 SWA gaps, 97.27%–99.95% of singular energy resides in a low-rank $r = 128$ subspace, restoring cosine alignment from $+0.0069 - +0.0320$ to $+0.7491 - +0.9527$. **(B)** Pulling $\{\texttt{L35}, \texttt{L41}\}$ into the canonical Vocabulary Fiber boosts 14-head 5-Hop accuracy to 87.1174%, collapses 5-Hop decoy error $21,163.8\times$ (95.2373% $\to$ 0.0045%), and raises 10-seed BABILong `qa3` mean target probability to 69.7890% (14.4656% decoy error).

Table 8: Phase 8 Stage 1B Low-Rank Orthogonal Fiber-Bundle Connection Diagnostics Across the 6 SWA Gaps ($d = 2,560$).

5-Layer SWA Gap	Raw Unrotated Cosine	Procrustes ($r = 128$) Cosine	Full SVD Cosine ($r = 2560$)	Top-16 SVD Energy	Top-128 SVD Energy	Mean Calib $\|\mathbf{u}_t\|_2$
L5 → L11	+0.0320	**+0.9149**	+0.9150	94.28%	**99.95%**	58.28
L11 → L17	+0.3433	**+0.9486**	+0.9513	95.19%	**99.62%**	54.38
L17 → L23	+0.0069	**+0.7912**	+0.8033	81.10%	**97.27%**	64.22
L23 → L29 (Phase Boundary)	+0.0249	**+0.9527**	+0.9535	95.63%	**99.49%**	**214.69** (3.7× Spike)
L29 → L35	+0.4231	**+0.7491**	+0.7491	92.19%	**99.72%**	54.25
L35 → L41	+0.4642	**+0.9008**	+0.9013	93.90%	**99.88%**	31.84

Table 9: Phase 8 Stage 2 Evaluation: Internal Procrustes Skip-Highway vs. Vocabulary-Fiber Pullback (`gemma-4-E4B-it`).

Configuration (`google/gemma-4-E4B-it`)	5-Hop Target (79421)	5-Hop Decoy (63819)	BABILong qa3 Top-1	BABILong qa3 Mean Target	BABILong qa3 Decoy
1. Unmodified Baseline	2.2398% (#2)	95.2373%	50.0% (5/10)	44.2188% ± 44.5990%	52.0598%
2. Horiz Cubic Q14 + SWA Shield (Phase 6/7 Ref)	76.6634% (#1)	5.5535%	**70.0%** (**7/10**)	68.1805% ± 42.5986%	17.0473%
3. Horiz Stepped RG Q28 (Phase 6/7 Ref)	**93.8942%** (**#1**)	0.9205%	50.0% (5/10)	53.2468% ± 42.0183%	22.7146%
4. Covariant Procrustes Highway ($Q = 14, \gamma_v = 0.12$)	28.5143% (#2)	60.3647%	60.0% (6/10)	54.0991% ± 41.4939%	31.6476%
5. Vocab-Fiber Pullback ($Q = 14$, Unsoftcapped 0.70L41 $+ 0.30$L35)	**87.1174%** (**#1**)	**0.0045%** (**21,163.8**× down)	**70.0%** (**7/10**)	**69.7890% ± 45.6708%**	14.7929%
6. Dual-State Vocab-Fiber Richardson ($Q = 14$, L35/41 − L23/29)	**85.4214%** (**#1**)	**0.0045%** (**21,163.8**× down)	**70.0%** (**7/10**)	**69.7563% ± 45.6520%**	**14.4656%** (**3.60**× down)
7. Full 2D Covariant (Procrustes + Dual Vocab Pullback, $Q = 14$)	32.3060% (#2)	67.5376%	60.0% (6/10)	53.6282% ± 44.3955%	25.8106%
8. Full 2D Stepped RG ($Q = 28$ + Procrustes + Dual Vocab Pullback)	35.5307% (#2)	64.3315%	60.0% (6/10)	45.6742% ± 38.8617%	26.9218%

While Procrustes alignment + unit-norm matching (`Config 4`) improves 5-Hop accuracy by 17.6× over Phase 6's unrotated skip highway (1.6152% → 28.5143%), linearly transporting Layer l_k's hop-m entity representation directly into Layer l_{k+1} partially bypasses the intervening nonlinear SWA+MLP lookup tables that advance the hop counter from $m \to m+1$. By contrast, **Vocabulary-Fiber Gauge Pullback (`Configs 5 & 6`)** leaves internal MLP hop transitions 100% intact while bypassing Gemma 4's 30 tanh(logits/30) softcapping ceiling, boosting 14-head 5-Hop accuracy from 76.6634% to **87.1174**%, collapsing 5-Hop decoy error **21,163.8**× (95.2373% → 0.0045%), raising 10-seed BABILong `qa3` mean target probability to **69.7890**% (+25.57% over Baseline), and cutting BABILong decoy error to **14.4656**%.

7 Summary of Patentable Claims Enabled by the Model

The theoretical model and empirical findings above support a comprehensive claim hierarchy for utility patent prosecution:

1. **Independent Claim 1 (Diagnostic Scanner via Precision-Matrix VIF & Spectral Mass):** Automated zero-shot identification of long-range causal driver heads in a pre-trained neural network by computing per-head empirical decay mass $\hat{\alpha}_{l,h} = (\bar{d}_{l,h} + \epsilon)^{-1}$ and the diagonal Variance Inflation Factors $\mathrm{diag}(\mathbf{R}^{+})$ of the inter-head entropy correlation matrix.

2. **Independent Claim 2 (Write/Read Asymmetric Query-Slice Extrapolation):** A runtime attention inference engine that preserves unmodified local gravity on document prefill rows ($i < T - W_q$) to protect intra-clause KV-cache entity binding while selectively activating spectral extrapolation on the active query/generation slice ($i \in [T - W_q, T)$), subject to token-0 sink pinning ($d_{i,0} = 0$) and a local syntactic guardband $d_{\text{long}}(i,j) = \max(0, i - j - W_{\text{guard}})$.

3. **Independent Claim 3 (Dual-State Symmetric Extrapolation with Iso-Potential Cubic Plateau):** Evaluating the attention distribution at symmetric positive and negative potential shifts $S_{\text{free}} = \mathrm{softmax}(L + \Delta L)$ and $S_{\text{heavy}} = \mathrm{softmax}(L - \Delta L)$ over a convex plateau potential $\Phi_l(i,j) = \min(1.0, d_{\text{long}}(i,j)/(c_l L))^p$ ($p \geq 2$), and combining them via $\mathcal{N}(\max(0, (1+\eta)S_{\text{free}} - \eta S_{\text{heavy}}))$ to simultaneously eliminate inter-zone cannibalization on the plateau ($d \geq c_l L$) and cancel mid/late-context decoys ($d < c_l L$).

4. **Dependent Claim 4 (Depth-Progressive Renormalization Group Relay & SWA Decoy Shielding):** Stratifying the selected driver heads across multiple Global Attention layers $\mathcal{G} = \{l_1, \ldots, l_K\}$ with monotonically increasing horizon cutoffs $c_{l_1} \leq c_{l_2} \leq \cdots \leq c_{l_K}$ (including the 7-step 5-Hop schedule $[0.38, 0.44, 0.52, 0.64, 0.74, 0.85, 0.90]$) coupled with optional SWA tail damping ($48 < d \leq 512$) to execute up to 5-hop relational composition across network depth in a single forward pass.